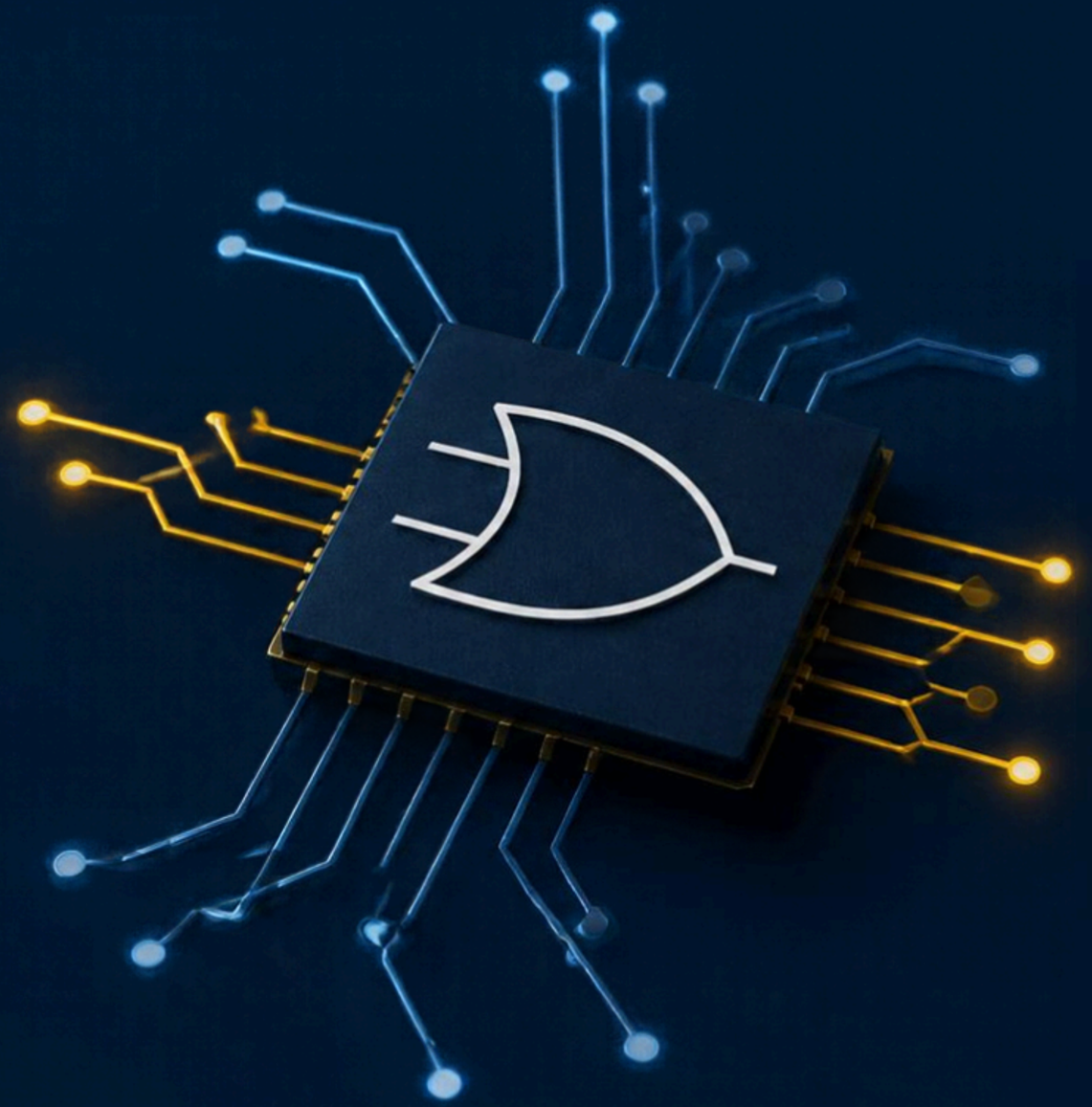
DESIGN BASIC AND UNIVERSAL LOGIC GATES

AND, OR, NOT, NAND, NOR, XOR, XNOR:
Working and Applications

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BASIC LOGIC GATES

FUNCTIONS AND APPLICATIONS

The building blocks of digital circuits.
Simple functions with
powerful real-world uses.



AND

Output true only
when all inputs are true



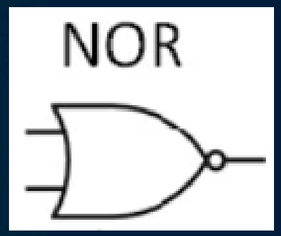
OR

Output true
if any input is true



NOT

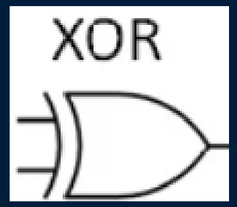
Inverts the
input signal





NAND AND NOR

Universal gates that can
implement all other gates





XOR AND XNOR

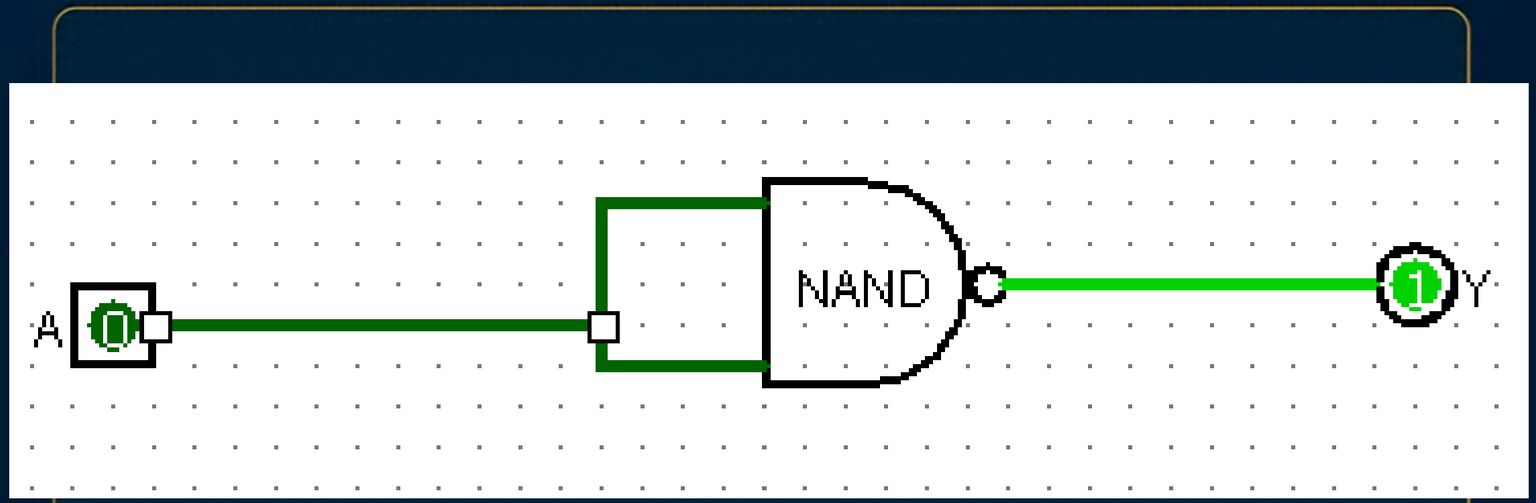
Used in arithmetic
and parity checks

NAND GATE AS A UNIVERSAL GATE AND INVERTER

NAND gates are universal and can implement any logical function

By tying both inputs together, a NAND gate becomes a **NOT gate**

The output is the **inverse** of the input

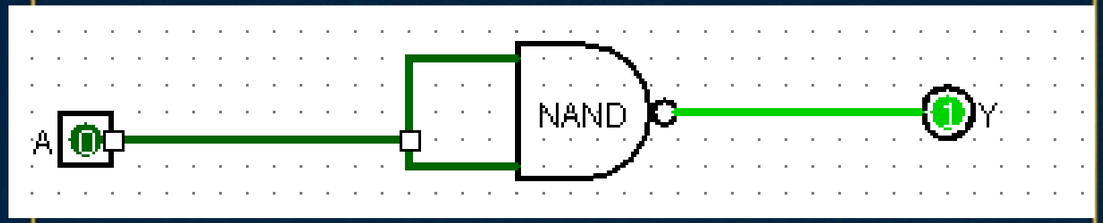


NAND gate used as NOT gate (Inverter)

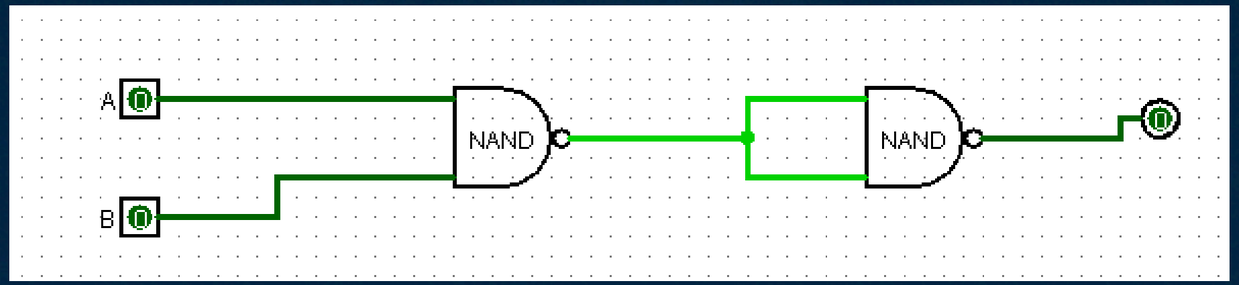
REALIZING AND, OR, AND NOT GATES
USING ONLY NAND

Combinations of NAND gates can replicate any basic logic function. These circuits use NAND gates alone to implement AND, OR, and NOT.

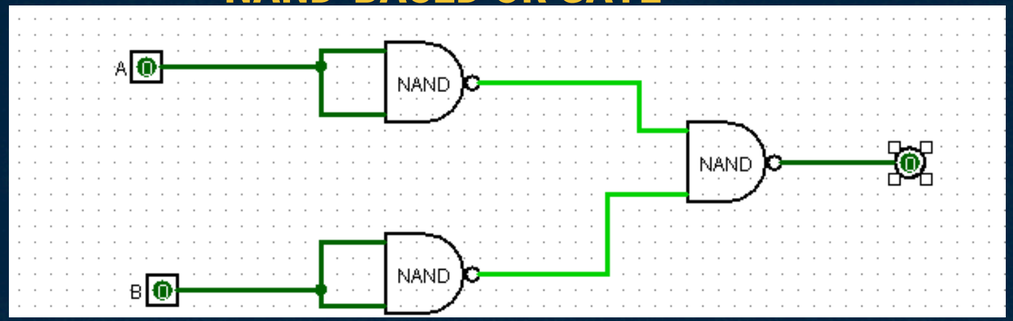
NAND-BASED NOT GATE



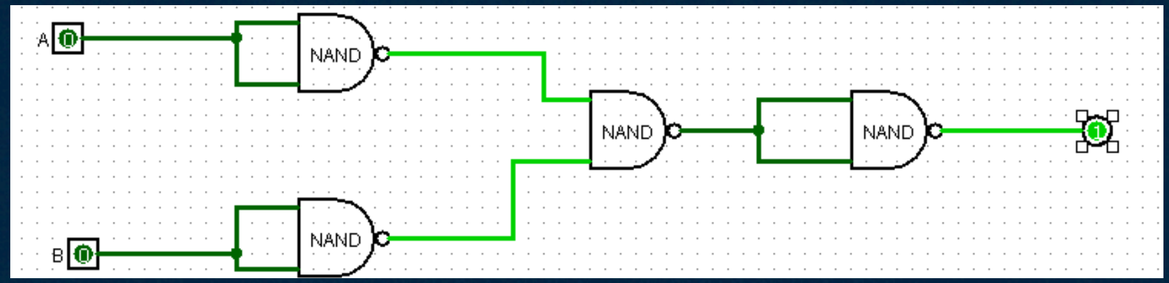
NAND-BASED AND GATE



NAND-BASED OR GATE

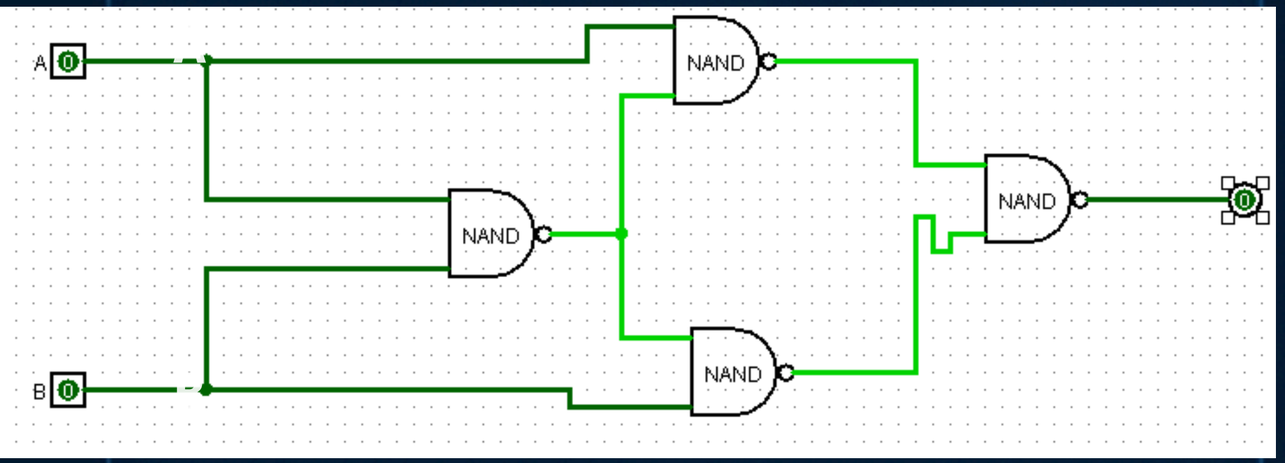


NAND BASED NOR GATE

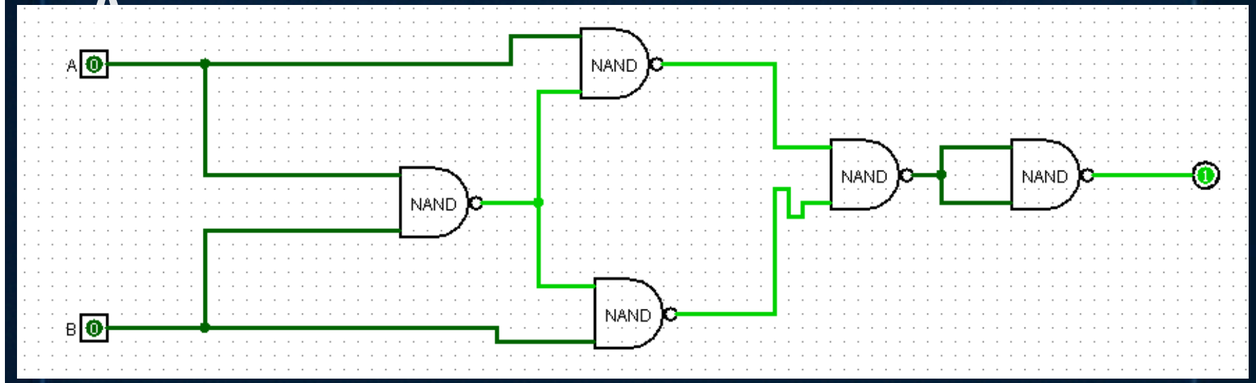


REALIZING XOR AND XNOR GATES USING ONLY NAND

XOR GATE USING ONLY NAND



XNOR GATE USING ONLY NAND



XOR and XNOR are essential in arithmetic circuits and parity generation.

They enable addition, subtraction, and comparison operations by identifying when inputs are different or the same.

SUMMARY AND APPLICATIONS OF

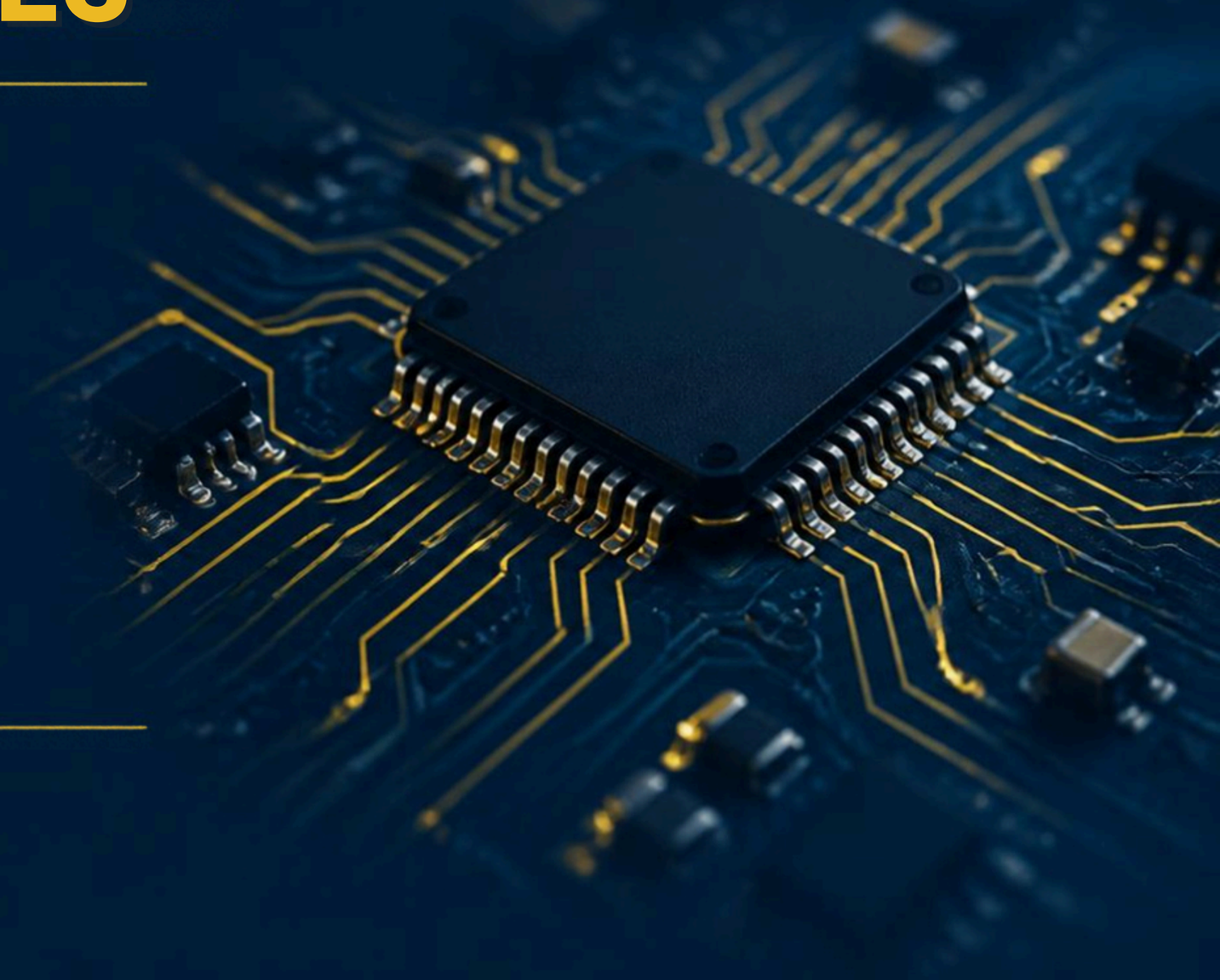
UNIVERSAL GATES

Universal gates simplify circuit design and manufacturing

NAND and NOR-based realizations provide design flexibility

Applied across computing, communications, and control systems

Explore NOR-only implementations for advanced designs



THANK YOU – QUESTIONS AND DISCUSSION WELCOME